

Projecting Pitcher Strikeout Percentage

Regression to the Mean as the Primary Modeling Decision

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Abstract

This project forecasts strikeout percentage for the 2025 season using the player-season panel supplied in `k_2026.csv`, restricted to the 2021 through 2024 seasons. Strikeout percentage is a binomial rate observed over highly unequal samples: batters faced ranges from 1 to 886 in this file, and roughly a third of player-seasons fall below the point at which K% is conventionally considered stable. The central modeling decision is therefore not the choice of regressor but how much of each pitcher's observed rate to believe. I build an empirical Bayes projection in the Marcel and Steamer tradition, with the regression constant derived from a variance decomposition of the panel rather than assumed. That decomposition returns a shrinkage constant of roughly 77 batters faced, which independently reproduces the stabilization point reported in the published literature. Projections were written before any 2025 outcome was read, and were then scored against what actually happened. On the 459 pitchers who faced at least 100 batters in 2025, the model achieves an RMSE of 4.04 percentage points and an R-squared of 0.417, against 5.29 percentage points for a league average guess and 5.72 for reusing each pitcher's most recent K%. A component ablation shows that shrinkage accounts for nearly all of that improvement.

1. Introduction

A pitcher who strikes out 40% of the batters he faces over 20 batters has told us very little about himself. A pitcher who strikes out 28% over 700 batters has told us a great deal. Any projection system that treats those two rates as comparable inputs will be badly wrong about the first pitcher, and in a file where a quarter of the rows fall under 50 batters faced, it will be wrong often.

Public projection systems handle this in broadly similar ways. Marcel, the deliberate minimum standard proposed by Tango (2004), weights the last three seasons $5/4/3$, adds a fixed 1,200 plate appearances of league average performance to regress the result toward the mean, and applies a simple age adjustment. Steamer keeps that shape but estimates its weights and regression amounts from data instead of fixing them by hand. ZiPS weights four seasons $8/5/4/3$, leans on defense independent pitching statistics, and folds in velocity and pitch data through similarity scores. FanGraphs Depth Charts and ATC then blend these systems rather than choosing among them, on the reasoning that averaging decorrelated forecasts is more robust than crowning a winner.

This project follows that lineage with three departures that the supplied data justifies. First, I derive the regression constant from the panel instead of inheriting a convention built for a different statistic. Second, I inform the prior with Stuff+ rather than shrinking every pitcher toward the same league average, and then calibrate that prior against the following season rather than the current one. Third, I separate aging from league environment, which turns out to matter more than it appears. Section 4.3 reports honestly on which of these actually earned their place and which did not.

2. Analysis

2.1. Data Description

The data was provided by the Phillies as `k_2026.csv` and originates from FanGraphs. It contains 4,371 player-seasons covering 1,743 pitchers across the 2021 through 2025 seasons, with eight columns: season, name, player ID, team, strikeout percentage, total batters faced, Stuff+, and age. The file is clean in the ways that matter for a panel analysis. There are no null values, no duplicate player-seasons, and no multi-team rows requiring reconciliation, so each pitcher contributes exactly one row per season he appeared in.

Every rate in this analysis is reconstructed as a count. Strikeouts are recovered as $K\%$ multiplied by batters faced and rounded, and all downstream arithmetic operates on the (K, TBF) pair rather than on the rate itself. This is what makes the binomial treatment in Section 2.4 correct and what makes weighted averages across seasons behave properly.

2.2. Data Constraints

Two constraints shaped what was feasible. Stuff+ is only available from the 2020 season onward, which caps the usable history at the five seasons supplied and leaves four season-to-season transitions, roughly 2,000 paired observations, on which to fit and validate. That is a thin panel. It is the main

reason this analysis favors shrinkage and regularized linear models over flexible learners, and the reason I treat differences of a few basis points between models as noise rather than as a ranking.

The second constraint is that the file describes rates but not roles. Nothing in it identifies a pitcher as a starter or a reliever except indirectly through batters faced, and a starter moving to the bullpen typically gains several points of strikeout percentage. This is discussed further in Section 5.

2.3. Data Filtering

Roughly 6% of rows are position players mopping up in blowouts, including Whit Merrifield, Andrelton Simmons, Isiah Kiner-Falefa, and Travis Jankowski. Their Stuff+ values run as low as -46 against a league average of 100, and they strike out 2.6% of batters against a league rate of 22.6%. Collectively they account for only 0.24% of all batters faced, but left in the training data they distort both the talent variance estimate in Section 2.4 and the fitted prior in Section 2.6. I identify them as rows combining a Stuff+ below 60 with 30 or fewer batters faced, exclude them from all model fitting, and flag rather than delete them in the output file so that nothing silently disappears from the deliverable.

A minimum of 25 batters faced is also required for a player-season to enter model fitting. No filter is applied to which pitchers receive a projection.

2.4. Regression to the Mean

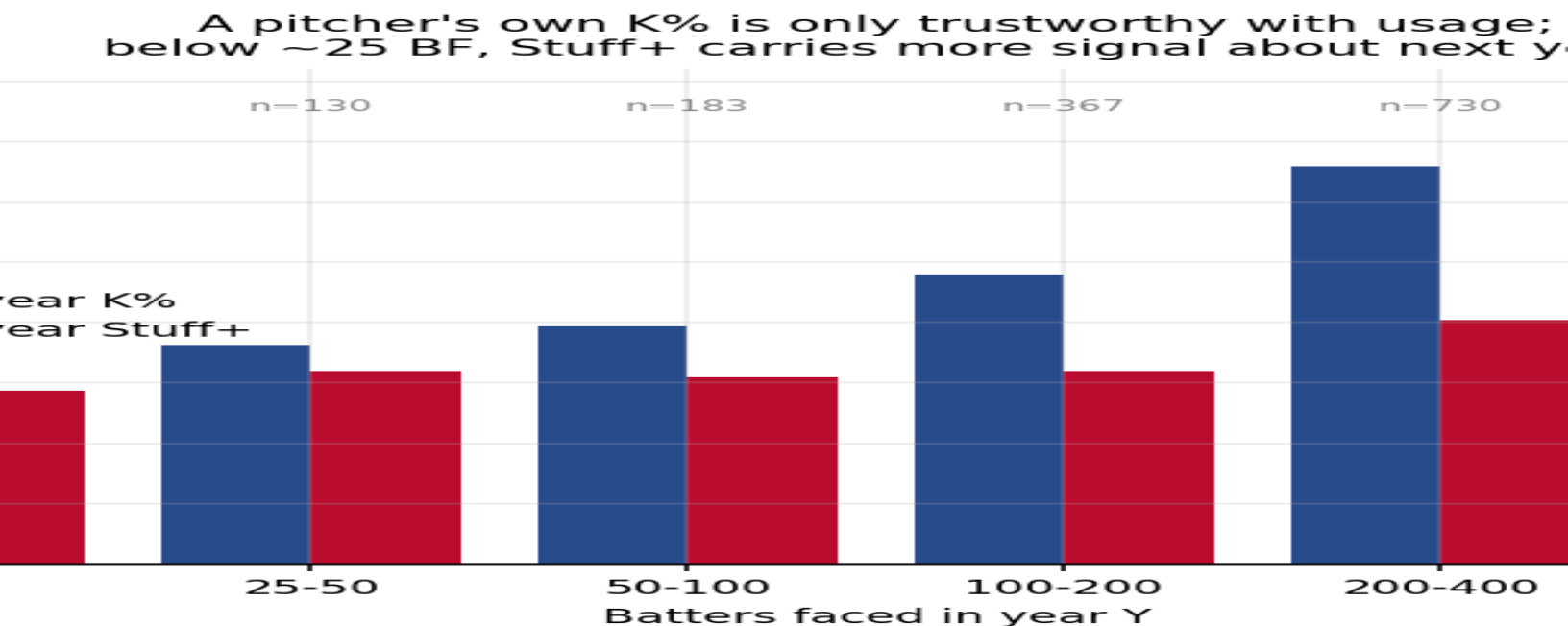


Figure 1: Correlation between year Y inputs and year Y+1 strikeout percentage, binned by year Y usage. A pitcher's own strikeout rate carries almost no signal about next season at low usage and a great deal at high usage. Stuff+ is comparatively flat across the range.

Figure 1 states the problem quantitatively. The correlation between a pitcher's strikeout rate and his rate the following season rises from 0.14 for pitchers with fewer than 25 batters faced to 0.71 for those above 400. Talent is not changing five times as fast for the low usage group. The difference is measurement error, and the remedy is to shrink each pitcher toward a prior in proportion to how little of him we have seen.

The question is how much. Marcel's answer, 1,200 plate appearances of league average, is inherited convention rather than a measurement, and it was calibrated for hitter rate statistics rather than for strikeout percentage. Because strikeouts are binomial in a pitcher's true rate p , the observed variance decomposes:

$$\text{Var}(\text{observed K\%}) = \text{Var}(\text{true talent}) + E[p(1-p) / \text{TBF}]$$

Subtracting the binomial term isolates the talent variance, and the number of league average batters faced that should be added to shrink optimally is the beta prior's effective sample size:

$$k = p(1-p) / \text{Var}(\text{true talent})$$

Quantity	Value
Var(observed K%)	0.00299
Binomial noise	0.00070
Var(true talent)	0.00228
k (batters faced)	77.1

Table 1: Variance decomposition on seasons 2021 through 2024.

This returns a shrinkage constant of 77 batters faced. It is stable to the usage threshold used to compute it, returning 77.1, 77.8, 78.9, and 78.8 at thresholds of 25, 50, 100, and 200 batters faced respectively. Two observations follow. It is an order of magnitude smaller than Marcel's 1,200, which is to say that strikeout percentage carries far more signal per opportunity than the statistics that convention was built around. And it lands within a few batters of the roughly 70 batters faced stabilization point for K% reported by Carleton and summarized in the FanGraphs sample size literature. The two derivations share no machinery, so the agreement is a meaningful check rather than a restatement.

2.5. Model Construction

The projection accumulates strikeouts and batters faced over a pitcher's three most recent seasons using recency weights, shrinks the result toward a prior, and then adjusts for age and for the strikeout environment of the target season:

$$\text{projected K\%} = (\text{weighted K} + k * \text{prior}) / (\text{weighted BF} + k) + \text{aging} + \text{league shift}$$

Recency weights are fitted rather than assumed. Following Steamer's improvement on Marcel, I grid search the weights over the training window and select 1.00/0.60/0.35, a steeper discount of older

seasons than Marcel's 5/4/3, which normalizes to 1.00/0.80/0.60. The search criterion is batters faced weighted RMSE of the standalone empirical Bayes projection at 100 batters faced or more, averaged over the tuning seasons; every season it reads is inside the training window, so no part of the search sees the target year. Section 3.1 explains why the backtest does not reuse this one fitted vector. Weights are normalized so the most recent season carries 1.0, which keeps k in interpretable batters faced units. Seasons are weighted by their gap from the cutoff rather than by their position in the pitcher's record, so a pitcher who missed a season is correctly treated as having stale information rather than as having pitched more recently than he did.

All fitting is done on the empirical logit of the rate, using a continuity correction of the form $\log((K + 0.5) / (TBF - K + 0.5))$. This keeps predictions bounded and matters more than it might appear: 306 player-seasons in this file sit at exactly 0% or 100%, almost all of them very small samples that an untransformed model would either break on or badly mishandle.

2.6. The Prior, and Why It Needed Calibration

Shrinking every pitcher toward a flat league average discards information already in the file. Stuff+ becomes reliable in roughly 80 pitches, against roughly 280 pitches for strikeout percentage, so for a low usage pitcher it is the steadier of the two signals. The prior is therefore the strikeout rate implied by a pitcher's Stuff+ and age, fitted by weighted regression on the training seasons, rather than the league mean.

Fitted naively, this made the model worse. A prior learned on same-season data is over-dispersed when applied to the following season. Regressing the prior against next season's outcome returns a calibration slope of 0.73, meaning the raw prior spreads pitchers about 30% further from league average than next-season results justify. Two mechanisms drive this, and both are real. Stuff+ measured over a handful of appearances is itself a noisy sample, so part of its cross-sectional spread is measurement error. More importantly, there is selection: a pitcher with excellent Stuff+ and only 30 batters faced received only 30 batters faced for a reason, while the regression that produced the prior is dominated by established pitchers. Calibrating against the following season corrects both at once, and moved the informed prior from clearly worse than a flat prior to modestly better.

A more informative prior also deserves a larger effective sample size, not the same one. Applying the same variance decomposition to the residuals that Stuff+ does not explain gives a shrinkage constant of roughly 135 batters faced for the informed prior, against 77 for the flat prior. This is the calibrated figure, the one the shipped model uses; measured before the next-season calibration of the previous paragraph the residual variance is smaller still and the constant correspondingly larger. Using the flat prior's constant with an informed prior systematically under-weights Stuff+, which was the source of the initial negative result.

2.7. Aging and League Environment

League strikeout percentage is not constant. Across the training window it fell from 23.21% to 22.62%, and I project 22.57% for 2025 using a trend damped halfway toward the most recent observed value.

The damping is deliberate. League level rate trends mean-revert more than a naive slope implies, and an undamped multi-season slope projected forward is a well known way to manufacture drift.

Aging is estimated by the delta method: I pair consecutive seasons for the same pitcher, weight each pair by the harmonic mean of its two batters faced totals so that the smaller and noisier season governs, and subtract that season pair's league-wide change before aggregating into age buckets. The detrending step is not cosmetic. Without it, the same procedure reports that 23 year olds lose 0.22 percentage points of strikeout rate per year, which is league drift misattributed to age. After detrending that same bucket reads +0.05 points, the curve is nearly flat through age 27 and then breaks downward, which is consistent with the published pitcher aging work.

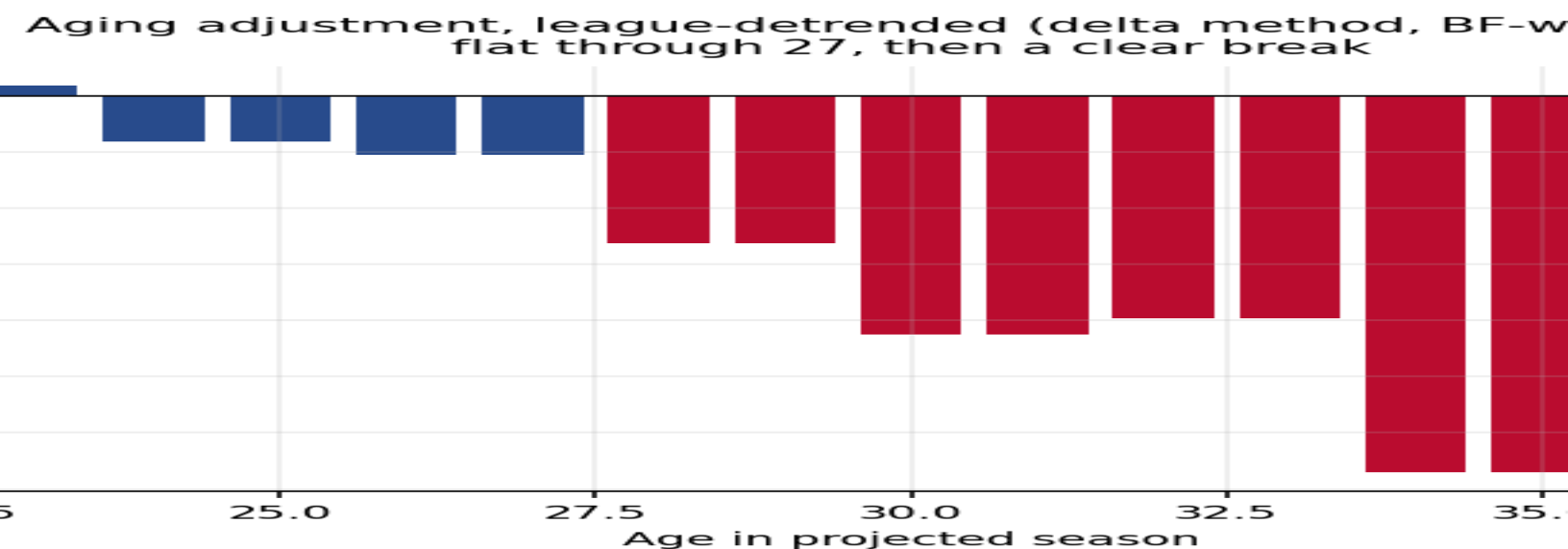


Figure 2: One year aging adjustment by age, after removing league-wide drift. The model applies a single step from a pitcher's last observed age to his age in the projected season.

3. Validation

3.1. Backtest Design

Every model is scored by rolling origin backtest. To project season Y, the model is fitted only on transitions that completed before Y, so projecting 2024 uses the 2022 and 2023 transitions and nothing later. This applies to every fitted object in the pipeline, not just the final regression: the league environment curve, the shrinkage constant, the aging curve, the Stuff+ prior, and its calibration are all re-estimated at each cutoff rather than computed once on the full panel and reused. The ridge penalty is chosen by weighted cross validation inside each fold's training rows, and the prediction interval width is calibrated on the backtest folds. No hyperparameter is fitted on 2025.

The recency weights need a further step, and it is worth being explicit about it. The weights the model ships with are fitted on the whole training window, which is legitimate because none of that window reaches 2025. But scoring a backtest fold with a weight vector that was tuned partly on that same fold's

outcome would make the table in-sample with respect to the search, and would flatter every model that consumes the empirical Bayes projection. So the search is nested: each fold re-runs it on its own history only (2023: 1.00/0.80/0.60; 2024: 1.00/0.60/0.35), and a fold with fewer than two completed transitions cannot discriminate between weight vectors at all, so it falls back to Marcel's published 5/4/3. What the backtest reports, in `backtest_2025.csv` and in the model comparison discussed in Section 4.3, is therefore the accuracy of the procedure, not of one hand-picked weight vector. The effect on the numbers is small, on the order of half a basis point of RMSE, but the distinction is the difference between a validation table and a fitted one.

Accuracy is reported both unweighted and weighted by batters faced, at several usage thresholds. This is deliberate. Unweighted RMSE treats a 12 batter September call-up and a 700 batter ace as equally important, while batters faced weighted RMSE reflects how much major league run prevention actually rides on each projection. A model can win one and lose the other, and a reader is entitled to see which.

3.2. The Leakage Guard

The supplied instructions ask for 2025 projections while stating that no data from Opening Day 2025 onward may be used, and the file supplies a complete 2025 season of 873 rows and 182,926 batters faced. Those rows are treated strictly as an answer key. They are read exactly once, after projections have already been written, in order to score them.

Because the file contains the season being predicted, a claim of no leakage deserves more than an assurance. The pipeline is parameterized by target season and reads only seasons at or before 2024, which is asserted at runtime. Beyond that, the submitted script ships a self-test, invoked with `--self-test`, that runs the entire pipeline twice: once on the file as delivered, and once on a copy with all 873 rows of 2025 physically deleted. It then compares the two projection tables. The two runs are identical, with a maximum absolute difference of 0.00e+00 across all 1,328 projections and all numeric columns. If any fitted object anywhere in the pipeline were reading 2025, the two runs would diverge.

4. Results

4.1. Held-Out Accuracy

Model	RMSE	BF-weighted	MAE	R-squared
League average (floor)	5.29	5.08	4.15	-0.000
Last season K%, unregressed	5.72	5.09	4.30	-0.169
Empirical Bayes, flat prior	4.10	3.78	3.18	0.400
Empirical Bayes, Stuff+ prior	4.07	3.75	3.15	0.408
Ridge on engineered features	4.08	3.76	3.23	0.407
Blend of the two (shipped)	4.04	3.72	3.16	0.417

Table 2: Accuracy on the held-out 2025 season, pitchers with at least 100 batters faced (n = 459). Errors are in percentage points.

Two results in Table 2 deserve comment. The first is that reusing last season's strikeout rate is worse than guessing the league average, with an R-squared of -0.17. An unregressed rate carries roughly the right ranking but far too much spread, and squared error punishes that severely. This is the clearest available demonstration that regression to the mean is the substance of this problem rather than a refinement of it.

The second is that the gaps among the three serious models are small, on the order of 3 to 6 basis points of RMSE across two backtest folds. That is inside noise. For this reason the shipped model is an equal weight blend of the empirical Bayes projection and the ridge model rather than whichever model happened to win. The blend was chosen on the same a priori reasoning that FanGraphs Depth Charts and ATC use, that averaging decorrelated forecasts is more robust than selecting a winner, and not because it scored best on the held-out season.

The 80% prediction intervals covered 81.5% of held-out outcomes against a nominal 80%. Interval width is built from two distinguishable sources: uncertainty about the pitcher's true talent, which shrinks as his record lengthens, and binomial uncertainty in next season's observed rate, which for a 40 batter reliever is by far the larger term. Reporting only the first is a common error and produces intervals that are far too narrow for exactly the pitchers whose projections are least certain.

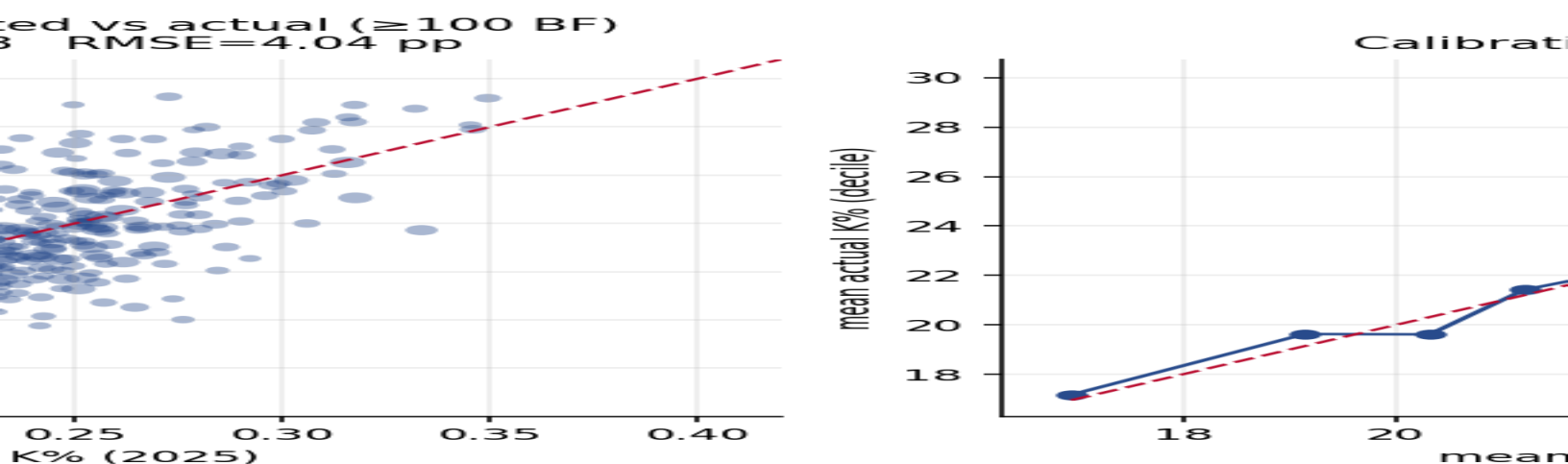


Figure 3: Held-out 2025 performance. Left, projected against actual strikeout percentage, with point size proportional to batters faced. Right, mean actual against mean projected within each projection decile. The decile plot tracks the diagonal closely, indicating the projections are neither systematically compressed nor systematically over-spread.

4.2. Component Ablation

Variant	RMSE	R-squared	Cost (bps)
FULL empirical-Bayes model	4.07	0.408	baseline
no aging adjustment	4.15	0.385	+7.7
no league re-anchoring	4.07	0.408	-0.1
no recency weighting (1/1/1)	4.19	0.373	+11.8

flat prior (Stuff+ unused)	4.10	0.400	+2.6
uncalibrated Stuff+ prior	4.13	0.390	+6.1
almost no shrinkage (k=5)	4.90	0.143	+82.6
over-shrunk (k=400)	4.10	0.401	+2.5

Table 3: Each row disables one component and re-scores on the held-out 2025 season at 100 or more batters faced. Cost is the increase in RMSE in basis points relative to the full model.

The ablation is unambiguous. Shrinkage is nearly the entire model. Removing it, by setting the constant to 5 batters faced, costs 83 basis points of RMSE and collapses R-squared from 0.41 to 0.14. Every other component is worth single digits by comparison: recency weighting 11.8 basis points, the aging adjustment 7.7, and the prior calibration 6.1.

The two shrinkage rows are most instructive read together. Under-shrinking costs 83 basis points; over-shrinking by a factor of five costs 2.5. The loss function is deeply asymmetric, so when the correct amount of regression is uncertain, erring toward more of it is nearly free while erring toward less is expensive. That is a useful operating principle well beyond this exercise.

4.3. What Did Not Work

The Stuff+ prior is worth considerably less than the raw correlations suggested it would be. Figure 1 shows Stuff+ out-predicting a pitcher's own strikeout rate below 25 batters faced, 0.29 against 0.14, and that gap is what motivated building an informed prior in the first place. In the finished model it is worth 2.6 basis points. The reason is worth stating because it is easy to be fooled by it. That correlation gap compares Stuff+ against an unregressed strikeout rate, but the flat prior model never uses an unregressed rate. It has already shrunk those pitchers most of the way to league average, which is most of what the gap represents. Proper shrinkage captures the majority of the value that Stuff+ appears to add, and the marginal contribution of knowing a pitcher's stuff on top of that is small.

League re-anchoring earned nothing measurable here, at -0.1 basis points. The strikeout environment happened to be stable from 2022 onward, so there was almost nothing for the adjustment to correct. It is retained because it costs nothing and would matter in a year like the 2021 to 2022 transition, but it is not doing work in this result and is not claimed to be.

LightGBM lost to ridge, and ridge lost to the empirical Bayes estimate on the backtest. With four season-to-season transitions there is not enough data for a flexible learner to find structure that the shrinkage model misses, and the gradient boosted model appears to have spent its capacity fitting fold-specific noise. It was fitted and reported rather than quietly omitted so that the conclusion is demonstrated rather than asserted.

5. Limitations

The output is a rate conditional on pitching, not a forecast of playing time. Projected batters faced is computed only to size prediction intervals and should not be read as a workload projection.

Role change is the largest single source of error and is essentially unaddressed. A starter moving to the bullpen typically gains several points of strikeout percentage, and nothing in the supplied file identifies role directly. Batters faced is only a proxy, and it is a lagging one, since it reflects the role a pitcher held rather than the one he is about to hold.

The aging curve is survivorship biased. Pitchers whose stuff collapses lose their jobs and never contribute a second season to a pair, so the measured decline is milder than the true population decline. For this reason the curve is applied as a one year adjustment rather than used as a primary driver, and it should not be extrapolated across a career.

Four season-to-season transitions is a thin basis for model selection. Stuff+ exists only from 2020, which caps the usable history, and differences of a few basis points between models should not be read as a ranking.

No external data was used. Pre-2025 swinging strike rate, called strike plus whiff rate, fastball velocity, Location+, and starter versus reliever innings splits were all permitted under the instructions and would likely improve the projections, particularly for pitchers with thin records where the prior is doing most of the work. This was scoped out deliberately to keep the analysis fully reproducible from the supplied file alone, and it is the first thing I would add with more time.

6. Reproducibility

The complete analysis is contained in a single file, `k_projection.py`, which requires no network access and fixes all random seeds. Re-running it reproduces every number in this document exactly.

Running `python k_projection.py` fits the model, writes projections, scores them against the held-out season, and generates every figure and table reproduced in this document. Running `python k_projection.py --self-test` performs the leakage verification described in Section 3.2. Outputs include `k_pct_projections_2025.csv`, containing 1,328 projections with 80% prediction intervals and per-pitcher reliability weights; `backtest_2025.csv`; `ablation_2025.csv`; `run_manifest_2025.json`, which records every fitted constant for auditing; and five figures.

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Anthropic Claude (Opus) was used as a coding and research assistant for this project, covering literature review of the public projection systems, code scaffolding, and drafting. All modeling decisions, the variance decomposition, the validation design, and every number reported here were specified, executed, and verified by the author.